1. Which of the following is true about 1NF?

(a) Allows multi-valued attributes
(b) Eliminates partial dependencies
(c) All attributes must be atomic
(d) Eliminates transitive dependencies

2. A relation is in 2NF if it is in 1NF and:

(a) Has no transitive dependencies
(b) Has no partial dependencies on primary key
(c) Every attribute is a prime attribute
(d) Has no multi-valued dependencies

3. Which normal form deals with multi-valued dependencies?

(a) 2NF
(b) 3NF
(c) BCNF
(d) 4NF

4. BCNF is a stricter version of:

(a) 1NF
(b) 2NF

(c) 3NF
(d) 4NF

5. Functional dependency X → Y means:

(a) Y determines X
(b) X determines Y
(c) X and Y are unrelated
(d) Both determine each other

6. Which of the following anomaly is NOT associated with unnormalized relations?

(a) Insertion anomaly
(b) Deletion anomaly
(c) Update anomaly
(d) Retrieval anomaly

7. A superkey is:

(a) A minimal set of attributes that uniquely identifies tuples
(b) Any set of attributes that uniquely identifies tuples
(c) An attribute that depends on a non-prime attribute
(d) A foreign key reference

8. If R(A,B,C) has FD: A → B and B → C, then C is:

(a) Partially dependent on A

(b) Fully dependent on A

(c) Transitively dependent on A

(d) Independently determined

9. 5NF is also known as:

(a) Domain-Key Normal Form

(b) Boyce-Codd Normal Form

(c) Project-Join Normal Form

(d) Element Normal Form

10. Lossless-join decomposition ensures:

(a) No data is lost during decomposition

(b) All FDs are preserved

(c) The relation is in BCNF

(d) No null values exist